

Smartphones and AI are remaking rural India

Villages have fallen in love with short-form videos and chatbots

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It is a warm afternoon in Nagepur and Bahadur Kumar is working at the power loom. Opposite the workshop stands his two-storey, half-finished house. Outside it are five goats, a motorbike, his wife and his 22-year-old son, Karan, who explains that he is busy doing a science degree at a local university while preparing for exams for a government job. Asked how he studies, he whips out a phone to show The Economist his favourite YouTube coaching classes and long conversations with Chatgpt about questions he does not understand.

Nagepur is a fairly well-off village by the standards of Uttar Pradesh, a huge and poor state. In the past decade many of its households have gained toilets, running water, reliable power—and the internet. Scarce not long ago, smartphones are becoming ubiquitous. Ownership remains skewed towards men, but devices are shared and keenly fought over. “When my dad goes to work, I take it,” says Roshni, a 14-year-old schoolgirl, flashing a smile.

This year the world’s most populous country is on track to pass 1bn active internet users, up from 622m in 2020, according to the Internet and Mobile Association of India (iamai), a trade body. The shift is happening at lurching speed thanks to dirt-cheap devices and data. Indians can now get their hands on a used Chinese-brand smartphone for as little as 2,000 rupees (\$21), around four days’ pay for a labourer. Data cost just 15 rupees (16 cents) per gigabyte, less than a tenth of the global average and much less than in most developing countries (see chart).

That is largely thanks to high population density, meaning that each mast can serve more customers. Data costs were also driven down by a brutal price war between Jio and Airtel, two telecoms behemoths (on June 19th Jio announced plans for a blockbuster initial public offering this year). Along with low rates of Wi-Fi installation, cheap tariffs help to explain why Indians are the hungriest consumers of mobile data in the world, according to Ericsson, a Swedish telecoms company.

The smartphone revolution has already swept Indian cities. As they approach saturation, the digital frontier has moved to the village. Internet usage is growing four times faster in rural areas than urban ones, according to iamai. It is surging above all among the young (like Karan and Roshni), promising to upend much about rural life, from education and work to leisure and love. Thus far Indian politicians have cheered on a booming digital economy. But they are starting to fret about harms.

Nagepur offers a window into how the change is playing out. At the bustling village school, smartphones have long been used for homework. But ai chatbots are a revelation. Ask a class of

11- to 14-year-olds whether they use them, and a wave of hands goes up. According to Sensor Tower, a data aggregator, ai apps have been downloaded more than 600m times in India. Among children who use phones for education, three-quarters report using ai for practice, doubt-solving and filling in gaps in knowledge.

This is not surprising when many Indian schools still offer a patchy education (the one in Nagepur is better than most). Several Indian ai companies are building online tutoring platforms. If ai can make ed-tech work better, few will have more to gain than knowledge-hungry Indian schoolchildren. For them, says Nandlal Master, the head teacher in Nagepur, the spread of phones and the rise of ai has “made information accessible”.

A few decades ago Nagepur was a village of weavers, but now most of the men work as labourers in nearby Varanasi. Access to technology helps youngsters like Karan hunt for skilled work farther afield. Women in the village say smartphones have given them more autonomy. Only around half of rural women have a mobile phone, but among younger ones ownership is rising fast. Some men have tried to prevent this, but they are fighting a losing battle. Last year 15 villages in the state of Rajasthan tried to ban married women from using smartphones, citing concerns about phone addiction (men, apparently, suffered no such problems). Amid a national outcry, the ban was withdrawn.

Beyond education and work, smartphones are changing Nagepur’s social geometry. It is more common now for the young to form friendships and (whisper it) relationships beyond the village. The rural young may, like their urban peers, start to demand a greater say in crucial decisions, like marriage. Some demographers think smartphones are already accelerating fertility decline in villages, because they expose rural women to the lifestyles of richer urban ones. If villagers become as screen-addicted as their urban peers, one might expect the spread of other social attitudes also to accelerate.

Mr Nandlal, for his part, worries that what is being spread most is not autonomy or ambition, but distraction. In Nagepur children admit that while they can use phones for learning, they spend most of their time watching reels or playing shooting games. Short-form video and algorithmic content are particularly hard to resist in sleepy villages. In Nagepur parents say their children are being “spoiled”. Several note that boys no longer want to do manual jobs. “The more they idle the more they watch the more they idle,” one grumbles.

Only around half of under-30s in India have a job, and many that do have too little work. Pradyuman, an underemployed taxi driver, complains that his phone bill has rocketed because of short-form videos: “When you are by yourself your phone is your best friend,” he says. Lacking better options, millions of young Indians are spending their time and energy trying to become content creators. They are “burning their fingers thinking that this is the way up,” says

Jaijit Bhattacharya of the Centre for Digital Economy Policy Research, a think-tank based in Delhi.

You will hear similar worries the world over. But Indians are already among the world's top screen-addicts, and all this suggests their screen use is only likely to grow. One response is satire: being jobless and chronically online is a source of pride for the Cockroach Janta Party, India's newest youth political movement, which has held protests in several cities this month. Anyone can be a member if they spend at least 11 hours a day online, the party says.

Regulators are taking a tougher line. In March Andhra Pradesh and Karnataka became the first states to ban children from using social media. Several others are poised to follow suit. Ashwini Vaishnaw, the it minister, has said that the national government is considering age-based social-media restrictions, noting that many countries have already taken tough action.

India will have to weigh any measures carefully. Shivam Jadaun, a lawyer, points out that blanket bans are an especially clumsy tool in a society where many devices are still shared. One risk is that parents respond by withdrawing access altogether, reducing the benefits as well as the harms. Many difficulties online stem from a failure to fix problems offline, from shoddy schools to a lack of jobs. It would be better, argues Mr Jadaun, to strengthen the regulation of platforms while taking steps to improve digital literacy.

On the steps of Karan's house, his mother, wrapped in a black sari and headscarf, has been listening intently to the conversation. She chips in to say that she approves of young people using technology to strive for a better life. Her only worry? That more of them, as a result, will want a "love marriage". ■

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